

Final Examination

HWRS 504 – Numerical Methods for Environmental Transport Problems

- Date: December 16, 2025
- Time: 2 PM – 4 PM
- Location: Harshbarger 203
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1. (12 points)

Provide brief but precise explanations to the following questions.

(a) The method of characteristics (MOC) transforms certain PDEs into ODEs. Explain geometrically (i.e., by drawing graphs) why MOC eliminates numerical dispersion for pure advection problems, e.g.,

$$\frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0.$$

(b) Explain the difference between cell-centered finite difference and finite volume methods in terms of how they enforce conservation and how they approximate spatial derivatives.

(c) Consider the diffusion equation $\partial u / \partial t = D \partial^2 u / \partial x^2$. Explain why forward Euler requires a severely restricted time step.

2. (28 points)

Consider the transient 1D advection–diffusion equation ($V > 0$)

$$\frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L.$$

We use the following numerical scheme (first-order upwind for advection, central for diffusion):

$$\frac{U_i^{n+1} - U_i^n}{\Delta t} + V \frac{U_i^n - U_{i-1}^n}{\Delta x} = D \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta x^2}.$$

- (a) Construct the modified equation up to $O(\Delta x)$ and identify the leading numerical diffusion term. Provide the expression for the numerical diffusion coefficient D_{num} .
- (b) Derive a stability condition using Fourier (von Neumann) stability analysis (you can assume an infinite or periodic domain for the von Neumann analysis). Express the condition in terms of the grid Peclet number and CFL number.
- (c) Explain why increasing the grid Peclet number causes oscillations for central-difference advection schemes but not for upwind schemes.

3. (20 points)

Consider the pure advection equation ($V > 0$)

$$\frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0.$$

For a uniform grid spacing, suppose we write the following three-point biased stencil:

$$\frac{\partial u}{\partial x} \approx aU_{i-2} + bU_{i-1} + cU_i.$$

- (a) Using Taylor expansion, determine conditions on a, b, c for first-order and second-order accuracy.
- (b) Using your result from part (a), explain why a linear three-point upwind stencil that is second-order accurate must include at least one coefficient that is negative.

(c) Give a qualitative explanation of why having a negative coefficient in such a stencil can lead to oscillations or wiggles in the numerical solution near sharp gradients. You can illustrate by constructing examples.

4. (20 points)

Consider the PINN formulation for the 1D steady-state diffusion equation:

$$\frac{d^2 u}{dx^2} = f(x), \quad u(0) = 0, \quad u(1) = 1.$$

The PINN loss is:

$$\mathcal{L} = \frac{1}{N_f} \sum_{j=1}^{N_f} (u''_{\theta}(x_j) - f(x_j))^2 + (u_{\theta}(0))^2 + (u_{\theta}(1) - 1)^2.$$

(a) Explain how automatic differentiation (AD) is used to obtain $u''_{\theta}(x_j)$ and why AD avoids the discretization errors of finite differences.

(b) For a single-hidden-layer neural network

$$u_{\theta}(x) = W_2 \sigma(W_1 x + b_1) + b_2,$$

where W_1 is the weight matrix from the input layer to the hidden layer, write the backpropagation expression for $\partial \mathcal{L} / \partial W_1$.

5. (20 points)

We consider a family of PDEs with variable coefficients:

$$\frac{d}{dx} \left(k(x) \frac{du}{dx} \right) = g(x), \quad 0 < x < 1,$$

with Dirichlet boundary conditions. Assume $k(x)$ and $g(x)$ vary across instances, and our goal is to learn the operator mapping:

$$(k, g) \mapsto u.$$

(a) Explain how a Neural Operator differs fundamentally from a standard neural network used for PINNs or supervised regression.

(b) For the Fourier Neural Operator (FNO), describe:

- how the Fourier transform is used in each layer,
- why global convolution in Fourier space can approximate PDE operators efficiently.

Appendix:

Equation(s) that may be useful for answering the questions.

Taylor series: Let $f(x)$ be C^n -continuous over the interval $[a, b]$, $f(x) \in C^n[a, b]$. Then for any points x_0 and x within $[a, b]$, we have

$$f(x) = f(x_0) + \frac{df}{dx} \Big|_{x_0} (x - x_0) + \frac{d^2f}{dx^2} \Big|_{x_0} \frac{(x - x_0)^2}{2!} + \frac{d^3f}{dx^3} \Big|_{x_0} \frac{(x - x_0)^3}{3!} + \dots$$

$$+ \frac{d^{n-1}f}{dx^{n-1}} \Big|_{x_0} \frac{(x - x_0)^{n-1}}{(n-1)!} + R_n, \quad (5)$$

where the remainder term R_n is

$$R_n \equiv \frac{d^n f}{dx^n} \Big|_{\xi} \frac{(x - x_0)^n}{n!}, \quad \xi \in [x_0, x]. \quad (6)$$